SOIL EROSION AND CONSERVATION · NOTE ST05

Stability of Gravity Dams and Retaining Walls — Forces, Middle-Third Rule, Safety Factors and Earth Pressure

Course topics: SWCE-503 T4, SWCE-503 T5

Dams and retaining walls

- Their structural design is nearly identical. A **dam** holds water upstream, provides storage and passes surplus flow over its crest; a **retaining wall** holds back earth, as in straight-drop spillways. Both must withstand earth pressure and water pressure — in masonry or concrete weirs, earth pressure comes mainly from upstream silt and water pressure from the water column, with different levels upstream and downstream.
- A gravity dam must not fail by **overturning, tension (rupture), sliding or crushing**.

Middle-third rule

- Masonry cannot take tension, so no tensile stress may develop: the centre of pressure — the resultant R — must lie within the **middle third of the base**. Example: base 6 m → between 2 m and 4 m from the heel.
- Then overturning is safe and tension is avoided, and crushing is safe if the masonry can carry twice the mean stress. Check with the reservoir both full and empty.

Forces on a unit length

Weight: $W = (a + b)/2 \times H \times \rho$ a = top width, b = bottom width, H = height, ρ = density of masonry ($2200\ kg/m^3$)

Centroid from the vertical face: $\bar{x} = (a^2 + ab + b^2)/[3(a + b)]$

Water pressure: $p = w \cdot x$; total $P = w \cdot h^2/2 \approx w \cdot H^2/2$ ($w = 1000\ kg/m^3$), acting at $h/3 \approx H/3$ above the base

Resultant: $R = \sqrt{(P^2 + W^2)}$

The notes' weight formula ends "× H ×" — the density ρ belongs at the end, as written here.

- **Equilibrium:** ΣV = 0, ΣH = 0, ΣM = 0 (moments about B), so R_v = W and R_H = P.

Moments about B: $P \times H/3 + W \cdot \bar{x} = R_v \cdot (\bar{x} + z) \rightarrow z = (P/W) \times H/3$ (the "shift of reaction" MN)

Eccentricity: $e = \bar{x} + z - b/2$

Safety checks

Check	Condition	Minimum factor of safety (as in the notes)
Overturning (about the downstream toe)	Overturning moment P × H/3 against balancing moment; FOS = (W × NA)/(P × H/3), NA = lever arm of W about the toe	1.2
Sliding	P ≤ μW (μ = coefficient of friction); FOS = μW/P. If short, deepen the cutoff or toe wall or roughen the base	1.5
Compression	f_max = (W/b)(1 + 6e/b) ≤ permissible stress	—
Tension	Resultant within the middle third: e ≤ b/6 (extreme limit BM = 2b/3)	—

$$\text{Base stresses: } f_{max} = (W/b)\cdot(1 + 6e/b);\ f_{min} = (W/b)\cdot(1 - 6e/b)$$

Illustration (not in the notes) — masonry weir, vertical upstream face, water to the top
a = 0.6 m, b = 2.0 m, H = 2.5 m, ρ = 2200 kg/m³, μ = 0.6
W = (0.6 + 2.0)/2 × 2.5 × 2200 = 7150 kg; P = 1000 × 2.5²/2 = 3125 kg
$\bar{x}$ = (0.36 + 1.2 + 4.0)/(3 × 2.6) = 0.713 m; z = (3125/7150) × 2.5/3 = 0.364 m
e = 0.713 + 0.364 − 1.0 = 0.077 m < b/6 = 0.333 m → no tension
f_max = 3575 × (1 + 0.231) ≈ 4400 kg/m²; f_min ≈ 2750 kg/m²
Overturning FOS = 7150 × (2.0 − 0.713)/(3125 × 0.833) ≈ 3.5 > 1.2
Sliding FOS = 0.6 × 7150/3125 ≈ 1.37 < 1.5 → deepen the cutoff/toe wall or roughen the base

Earth pressure

Dry and wet earth pressure both affect stability. The **angle of repose** — the steepest slope loose earth stands at on a horizontal plane — is also called the angle of internal friction.

Soil	Angle of repose — dry	Angle of repose — wet	Unit weight (g/cc)
Sand	30–37°	38°	1.45–1.90
Loamy soil	30°	45–49°	1.60–1.92
Clay	30°	45°	1.92–2.20
Gravel	48°	48°	1.45–1.75

$$\text{Rankine earth pressure (unit length): } P = (W_e\cdot H^2/2)\cdot(1 - \sin\theta)/(1 + \sin\theta)$$

W_e = density of earth; θ = angle of repose; wall weight, centroid, eccentricity and base stresses as above

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. No tension develops in a masonry dam if the resultant lies within

(a) the middle half
(b) the middle third
(c) the outer third
(d) the heel

Q2. For a base width of 6 m, the middle third lies between ___ from the heel

(a) 0 and 2 m
(b) 2 and 4 m
(c) 3 and 6 m
(d) 1 and 5 m

Q3. Total water pressure on a vertical face of height H acts at

(a) H/2
(b) H/3 above the base
(c) 2H/3 above the base
(d) the top

Q4. Factor of safety against sliding is

(a) μW/P
(b) P/μW
(c) W/P
(d) μP/W

Q5. Minimum factor of safety against sliding (as in the notes) is

(a) 1.0
(b) 1.2
(c) 1.5
(d) 3.0

Q6. Maximum base stress is

(a) W/b
(b) (W/b)(1 + 6e/b)
(c) (W/b)(1 - 6e/b)
(d) 6We/b²

Q7. Rankine earth-pressure coefficient is

(a) (1 + sinθ)/(1 - sinθ)
(b) (1 - sinθ)/(1 + sinθ)
(c) tanθ
(d) sinθ

Q8. Angle of repose of dry sand is about

(a) 15–20°
(b) 30–37°
(c) 48°
(d) 60°

Answers

Q1 — (b) the middle third
Q2 — (b) 2 and 4 m
Q3 — (b) H/3 above the base
Q4 — (a) μW/P
Q5 — (c) 1.5
Q6 — (b) (W/b)(1 + 6e/b)
Q7 — (b) (1 - sinθ)/(1 + sinθ)
Q8 — (b) 30–37°